



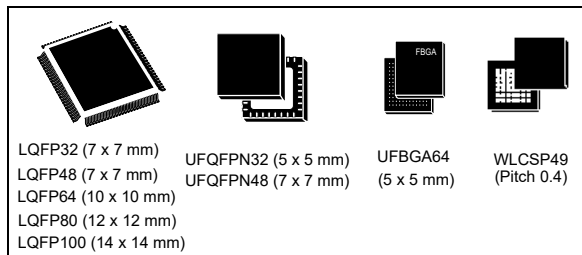
# STM32G431x6 STM32G431x8 STM32G431xB

Arm<sup>®</sup> Cortex<sup>®</sup>-M4 32-bit MCU+FPU, 170 MHz / 213 DMIPS,  
up to 128 KB Flash, 32 KB SRAM, rich analog, math accelerator

Datasheet - production data

## Features

- Core: Arm<sup>®</sup> 32-bit Cortex<sup>®</sup>-M4 CPU with FPU, Adaptive real-time accelerator (ART Accelerator) allowing 0-wait-state execution from Flash memory, frequency up to 170 MHz with 213 DMIPS, MPU, DSP instructions
- Operating conditions:
  - $V_{DD}$ ,  $V_{DDA}$  voltage range: 1.71 V to 3.6 V
- Mathematical hardware accelerators
  - CORDIC for trigonometric functions acceleration
  - FMAC: filter mathematical accelerator
- Memories
  - 128 Kbytes of Flash memory with ECC support, proprietary code readout protection (PCROP), securable memory area, 1 Kbyte OTP
  - 22 Kbytes of SRAM, with hardware parity check implemented on the first 16 Kbytes
  - Routine booster: 10 Kbytes of SRAM on instruction and data bus, with hardware parity check (CCM SRAM)
- Reset and supply management
  - Power-on/power-down reset (POR/PDR/BOR)
  - Programmable voltage detector (PVD)
  - Low-power modes: sleep, stop, standby and shutdown
  - $V_{BAT}$  supply for RTC and backup registers
- Clock management
  - 4 to 48 MHz crystal oscillator
  - 32 kHz oscillator with calibration
  - Internal 16 MHz RC with PLL option ( $\pm 1\%$ )
  - Internal 32 kHz RC oscillator ( $\pm 5\%$ )
- Up to 86 fast I/Os
  - All mappable on external interrupt vectors
  - Several I/Os with 5 V tolerant capability



- Interconnect matrix
- 12-channel DMA controller
- 2 x ADCs 0.25  $\mu$ s (up to 23 channels). Resolution up to 16-bit with hardware oversampling, 0 to 3.6 V conversion range
- 4 x 12-bit DAC channels
  - 2 x buffered external channels 1 MSPS
  - 2 x unbuffered internal channels 15 MSPS
- 4 x ultra-fast rail-to-rail analog comparators
- 3 x operational amplifiers that can be used in PGA mode, all terminals accessible
- Internal voltage reference buffer (VREFBUF) supporting three output voltages (2.048 V, 2.5 V, 2.95 V)
- 14 timers:
  - 1 x 32-bit timer and 2 x 16-bit timers with up to four IC/OC/PWM or pulse counter and quadrature (incremental) encoder input
  - 2 x 16-bit 8-channel advanced motor control timers, with up to 8 x PWM channels, dead time generation and emergency stop
  - 1 x 16-bit timer with 2 x IC/OCs, one OCN/PWM, dead time generation and emergency stop
  - 2 x 16-bit timers with IC/OC/OCN/PWM, dead time generation and emergency stop
  - 2 x watchdog timers (independent, window)
  - 1 x SysTick timer: 24-bit downcounter
  - 2 x 16-bit basic timers
  - 1 x low-power timer